

# Investigating the Effect of Model Complexity on Generalization Using Polynomial Regression

Team 01 | 24SJPCST503 – MACHINE LEARNING  
Course Project Report — Machine Learning Investigation Challenge

## 1. Team Members

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## 2. Research Question

**When does increasing model complexity stop improving generalization?**

Choosing how flexible a model should be is one of the central decisions in machine learning. A more flexible model can always fit its training data at least as well as a simpler one, so training error alone can never show when extra flexibility has stopped being useful. What matters is performance on data the model has not seen. By examining where training and unseen-data performance separate, this investigation looks at how much complexity a model should be given and why it has to be judged on held-out data.

## 3. Hypothesis

We expected that increasing the polynomial degree would first improve performance, because a more flexible model can capture more of the relationships between the physicochemical measurements and wine quality, and training error should keep decreasing as flexibility grows. We further expected that beyond some appropriate level of complexity, validation performance would stop improving, since the model would begin to fit patterns specific to the training sample rather than relationships that carry over to new data. Further complexity was therefore expected to cause overfitting and a deterioration in generalization, visible as a widening gap between training and validation error.

## 4. Theoretical Background

**Model complexity and polynomial regression.** Model complexity is the range of relationships a model is able to represent. Here it is controlled through the degree of a polynomial expansion: PolynomialFeatures replaces the 11 input variables by all their powers and interaction terms up to the chosen degree, and linear regression then fits one weight per expanded feature. The model stays linear in its parameters, but as a function of the original inputs it becomes steadily more flexible, and its number of parameters grows very quickly with the degree.

**Underfitting, overfitting and generalization.** A model that is too rigid underfits: it misses real structure and shows high error on both training and unseen data. A model that is too flexible overfits: it also absorbs variation that belongs only to the training sample, so training error keeps falling while error on unseen data rises. Generalization is performance on unseen data, and we quantify the loss in moving from seen to unseen data by the generalization gap (validation RMSE minus training RMSE).

**Bias–variance trade-off.** Error on unseen data reflects bias (systematic error from a model that is too rigid), variance (sensitivity to the particular training sample) and irreducible noise. Raising complexity lowers bias but raises variance. The relationship we test can therefore be summarised as:

- **Low complexity** may underfit (high bias);

- **moderate complexity** can capture useful patterns;
- **excessive complexity** may overfit (high variance);
- **overfitting** appears as improving training performance with deteriorating unseen-data performance.

## 5. Dataset / Data Source

The investigation uses the UCI Wine Quality dataset, red wine subset (Cortez et al., 2009), loaded directly from the UCI repository. It contains **1,599 observations**, **11 physicochemical input features** and one target, **wine quality**. The features are fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, free sulfur dioxide, total sulfur dioxide, density, pH, sulphates and alcohol. There are **no missing values**.

The target takes integer values from 3 to 8 (mean 5.64, standard deviation 0.81), and is concentrated around the middle scores: 681 wines have quality 5 and 638 have quality 6 (together about 82% of the data), whereas quality 3, 4 and 8 have only 10, 53 and 18 wines. The dataset suits this study because the prediction task, features and target stay identical while only the polynomial degree is changed, which allows model complexity to be investigated in a controlled way. Its uneven representation of quality values is a limitation (Section 9). The study examines model behaviour; it is not intended as a production wine-quality system.

## 6. Experimental Design

The 11 features were separated from the target, and the data was split randomly into **70% training (1,119 rows)**, **15% validation (240 rows)** and **15% test (240 rows)** with random state 42. For every degree we used the same pipeline: standardization of the inputs, PolynomialFeatures (no bias column), standardization of the expanded features, and ordinary least-squares LinearRegression with no regularization. Keeping all preprocessing inside the pipeline means that scaling statistics are learned from training data only, so no validation or test information leaks into training.

**Polynomial degree is the only experimental variable** (degrees 1 to 5, exceeding the required minimum of four levels); the dataset, split, preprocessing, algorithm and metrics are held constant. Several levels are needed because a single model cannot show a trend: only by comparing complexity levels side by side can we see where validation performance turns from improvement to deterioration. **RMSE** is the primary metric, with **MAE** and **R<sup>2</sup>** as supporting metrics, each computed separately on training and validation data. Validation performance was used to select the complexity level; the test set was kept separate and used once, for the final evaluation of the selected model.

**Table 1:** Polynomial degree versus model complexity.

| Degree | Number of polynomial features | Exceeds the 1,119 training rows? |
|--------|-------------------------------|----------------------------------|
| 1      | 11                            | No                               |
| 2      | 77                            | No                               |
| 3      | 363                           | No                               |
| 4      | 1,364                         | Yes                              |
| 5      | 4,367                         | Yes                              |

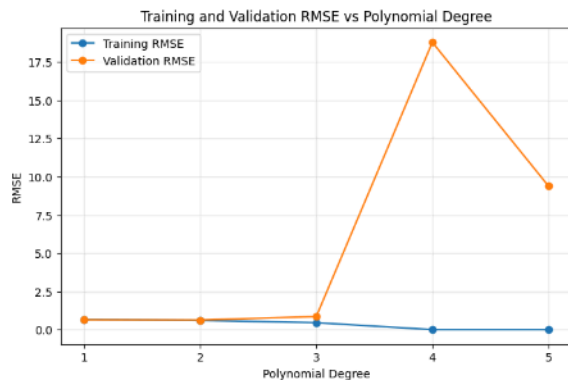
## 7. Results

Table 2 gives the training and validation performance for each degree, as computed in the notebook. Figure 1 plots the RMSE values and Figure 2 the resulting generalization gap.

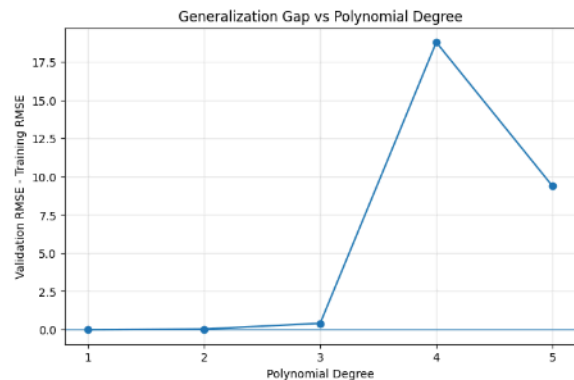
**Table 2:** Training and validation performance across polynomial degrees (primary split).

| Degree | Train RMSE             | Val. RMSE     | Train MAE              | Val. MAE | Train R <sup>2</sup> | Val. R <sup>2</sup> |
|--------|------------------------|---------------|------------------------|----------|----------------------|---------------------|
| 1      | 0.6487                 | 0.6374        | 0.4952                 | 0.5184   | 0.3612               | 0.3180              |
| 2      | 0.6015                 | <b>0.6274</b> | 0.4643                 | 0.5153   | 0.4508               | 0.3392              |
| 3      | 0.4544                 | 0.8563        | 0.3403                 | 0.6178   | 0.6865               | −0.2308             |
| 4      | $1.34 \times 10^{-13}$ | 18.7897       | $7.13 \times 10^{-14}$ | 6.5497   | 1.0000               | −591.5555           |
| 5      | $3.50 \times 10^{-14}$ | 9.3937        | $1.88 \times 10^{-14}$ | 3.6224   | 1.0000               | −147.1036           |

Values as reported in the notebook (rounded). Bold: lowest validation RMSE.



**Figure 1:** Training and Validation RMSE vs Polynomial Degree (notebook output).



**Figure 2:** Generalization gap (validation RMSE – training RMSE) vs polynomial degree (notebook output).

**Training performance improved with every increase in degree:** training RMSE fell from 0.6487 (degree 1) to 0.6015, 0.4544, and then to effectively zero (about  $10^{-13}$ ) at degrees 4 and 5, where training R<sup>2</sup> reaches 1.0. **Validation performance did not follow.** It improved from degree 1 to degree 2 (RMSE 0.6374 → 0.6274; R<sup>2</sup> 0.3180 → 0.3392), so degree 2 gave the lowest validation RMSE in the primary experiment. At degree 3 the validation RMSE rose to 0.8563 (R<sup>2</sup> −0.2308) although training RMSE kept falling, and at degrees 4 and 5 it reached 18.7897 and 9.3937, with strongly negative validation R<sup>2</sup>. The RMSE gap grew from −0.0112 (degree 1) and 0.0260 (degree 2) to 0.4019 (degree 3), 18.7897 (degree 4) and 9.3937 (degree 5).

The degree-2 model, selected on validation performance, was evaluated once on the untouched test set and gave **Test RMSE = 0.6548, Test MAE = 0.5078 and Test R<sup>2</sup> = 0.3621.**

**Robustness check.** To see whether the pattern depends on a single split, the degree comparison was repeated on three further random 80/20 train/validation splits (seeds 1, 42 and 100; this check has no test set and differs from the primary 70/15/15 split). Table 3 shows the validation RMSE values.

**Table 3:** Validation RMSE across three random 80/20 splits (robustness check).

| Degree | Seed 1  | Seed 42  | Seed 100 | Mean    | Std. dev. |
|--------|---------|----------|----------|---------|-----------|
| 1      | 0.6189  | 0.6245   | 0.6544   | 0.6326  | 0.0191    |
| 2      | 0.6318  | 0.6179   | 0.6458   | 0.6319  | 0.0139    |
| 3      | 1.2037  | 0.8101   | 0.9058   | 0.9732  | 0.2053    |
| 4      | 20.2764 | 133.7479 | 39.3728  | 64.4657 | 60.7557   |
| 5      | 11.3486 | 19.3037  | 11.3741  | 14.0088 | 4.5855    |

Degrees 1 and 2 give nearly identical mean validation RMSE (0.6326 and 0.6319); degree 1 was lower in one split (seed 1) and degree 2 in the other two. Degrees 3, 4 and 5 were substantially worse than both in every split.

## 8. Analysis and Interpretation

**Why training error keeps falling.** Each degree's feature set contains all the lower-order terms of the previous one, and least squares chooses the weights that minimise training error over the richer set. Training error therefore cannot increase with degree; its decline ( $0.6487 \rightarrow 0.6015 \rightarrow 0.4544 \rightarrow \approx 0$ ) is guaranteed by construction and says nothing about generalization. This is why the model with the best training performance, degrees 4 and 5 here, cannot be regarded as the best model.

**Why moderate complexity helps, and only a little.** Moving from degree 1 to 2 added 66 features ( $11 \rightarrow 77$ ) and improved both training and validation performance, while the gap stayed small ( $-0.0112$  to  $0.0260$ ). This suggests that a purely linear model is slightly too rigid for this task and that some quadratic or interaction structure is real and transferable. The improvement is small, however ( $0.010$  in validation RMSE), and even degree 2 explains only about a third of the variance in validation quality ( $R^2 = 0.3392$ ). Degree 1 therefore shows at most mild underfitting; its similar training and validation errors ( $R^2$   $0.3612$  and  $0.3180$ ) more likely reflect limited signal in the features than a badly mis-specified model.

**Where improvement stops.** At degree 3 the direction reverses: 286 further features ( $363 - 77$ ) lowered training RMSE by about  $0.147$  but raised validation RMSE by about  $0.229$ , and validation  $R^2$  became negative, meaning predictions were worse than simply using the mean of the validation targets. The additional flexibility was being spent on training-specific variation. With 363 free weights fitted to 1,119 rows, and no regularization to restrain them, the model has ample freedom to do so. This is the clearest evidence in the experiment that generalization stopped improving between degrees 2 and 3, and the widening gap ( $0.4019$ ) is the direct signature of overfitting.

**Degrees 4 and 5: overfitting plus numerical instability.** From degree 4 onward the number of features ( $1,364$  and  $4,367$ ) exceeds the number of training rows ( $1,119$ ), so the least-squares problem can be solved exactly and the model can reproduce every training target, including its noise. The training RMSE of about  $10^{-13}$  is floating-point zero rather than a genuine level of accuracy. Validation RMSE values of  $18.79$  and  $9.39$ , roughly 30 and 15 times the degree-2 value, are also far larger than the spread of the target itself (standard deviation about  $0.81$  on a scale of 3 to 8), so these models are not merely slightly inaccurate; their predictions are wildly wrong on unseen points. We do not attribute this solely to overfitting. High powers of the same 11 inputs vary together, which makes the fitted weights very unstable, and raising standardized inputs to the fourth or fifth power greatly amplifies any unusual value; the summary statistics show long tails in some features (for example, total sulfur dioxide has a maximum of 289 against a mean of 46). Overfitting and the ill-conditioning of a very high-dimensional expansion probably act together, and this experiment did not separate them. Degree 5 having a lower validation RMSE than degree 4 ( $9.39$  versus  $18.79$ , and in all three robustness splits) should not be read as better generalization: both are catastrophic, and the size of these errors varies enormously between splits (degree 4 ranged from  $20.28$  to  $133.75$ ). A plausible but untested explanation is that degree 4, with 1,364 features against 1,119 rows, lies closer to the point where exact interpolation first becomes possible, where the fitted solution is most unstable.

**Link to the bias–variance trade-off.** The pattern is consistent with the trade-off. The step from degree 1 to 2 behaves like a bias reduction, since both errors fall. Beyond degree 2, variance dominates: the standard deviation of validation RMSE across splits (Table 3) rises from  $0.0191$  and  $0.0139$  for degrees 1 and 2 to  $0.2053$  for degree 3,  $60.7557$  for degree 4 and  $4.5855$  for degree 5, showing that highly flexible models change drastically with the particular data they see. This is supporting evidence rather than a formal bias–variance decomposition, which was not performed.

**Interpreting the best degree carefully.** Degree 2 produced the lowest validation RMSE in the primary experiment. However, the robustness analysis showed that Degrees 1 and 2 had very similar mean validation RMSE values, so the exact distinction between them is less certain. In contrast, Degrees 3–5 consistently showed substantially poorer generalization. The test result supports this reading: the degree-2 test RMSE

(0.6548) is close to its validation RMSE (0.6274), so selecting on validation data did not produce an unreasonably optimistic estimate, but the modest  $R^2$  (0.3621) shows that the model captures only part of what determines wine quality.

## 9. Critical Evaluation and Limitations

**Dependence on the split.** The primary experiment uses one fixed 70/15/15 split with a validation set of 240 rows. The exact numbers, and in particular which of degrees 1 and 2 wins, depend on which rows fall in each set; the robustness check confirms this. The check itself has only three splits and does not use cross-validation, so it is indicative rather than conclusive. What the experiment can establish is the qualitative pattern (degrees 3–5 generalize worse), not a precise optimal degree.

**Rapid growth of polynomial features.** Complexity in this design is tied to feature count, which rises from 11 to 4,367. At degrees 4 and 5 there are more features than training rows and no regularization is used, so complexity, dimensionality and numerical instability are confounded (Section 8). The results show that these high-degree models fail; they do not show how much of the failure is due to overfitting alone.

**Uneven target distribution.** About 82% of wines have quality 5 or 6, while extreme scores are rare, so the models see very few examples of low and high quality and the errors mainly reflect performance on typical wines. Behaviour may differ on data with a more balanced target.

**Single dataset and model family.** Only red-wine data and unregularized polynomial regression were studied. The conclusion that added complexity stops helping should not be assumed to transfer to other datasets or algorithms, and regularization, more data or a different notion of complexity could move the point at which benefits end. The experiment shows this behaviour clearly for one controlled case; it does not establish a general rule about the best degree.

## 10. Conclusion

Increasing model complexity does not continuously improve generalization. In this experiment, validation performance was best at **Degree 2** in the primary split (validation RMSE 0.6274), with a test RMSE of 0.6548 for the selected model; beyond that level it deteriorated, with the degree-3 model showing a clear rise in validation error and degrees 4 and 5 failing severely despite near-perfect training fit.

**What happened?** Training error fell at every step in complexity, but validation error improved only from degree 1 to 2 and then rose, sharply at degrees 4 and 5, while the training–validation gap grew from about 0.03 to 18.79 RMSE units.

**Why did it happen?** Added flexibility can always lower training error, but beyond a modest level the extra terms fit sample-specific variation rather than transferable structure. With unregularized least squares, 1,119 training rows and up to 4,367 features, the higher-degree models reproduce the training data almost exactly and become unstable on new data, an outcome consistent with overfitting and high variance, compounded by numerical instability.

**How confidently can we conclude this?** We are confident that low complexity generalizes much better than high complexity here: the robustness check strengthens this, as degrees 3–5 were substantially worse in every split. We are much less certain about the exact distinction between degrees 1 and 2, whose mean validation RMSE values are almost identical (0.6326 and 0.6319). The conclusion applies to this dataset and this unregularized polynomial setup, and not necessarily to other datasets or algorithms.

## 11. AI Usage Declaration

**AI tool(s) used:** ChatGPT, Claude, Gemini

**Purpose of using AI:** Understanding ML concepts, structuring the investigation, generating/debugging code, improving documentation and report writing.

**Parts of the project assisted by AI:** Conceptual explanation, experiment planning, coding assistance, debugging, visualization/documentation.

**How the AI-generated output was verified:** All code was executed and checked, and numerical results were verified against the actual notebook outputs.

**One AI suggestion that we modified, rejected, or corrected:** AI-generated suggestions were reviewed against the project requirements and actual experimental results; unsupported or unnecessary suggestions were not included.

## 12. References

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<https://github.com/joegeorge022/ML-Project>